Osptek Display

TFT LCD SPECIFICATION

Model No:

YDP400B001-V1

Revision Record 修改记录

Rev No.版本号	Date 时间	Description 内容
V.0	2023-03-27	首次发行
V.0	2023-03-27	修改背光厚度及驱动
V.0	2023-03-27	修改排线外形加 EMI
V.0	2023-03-27	增加高温胶

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1.General Specifications 基本规格

No.	Item 项目	Specification 规格	Unit 单位	Remark
1	LCD Size 液晶面板尺寸	4.0''	inch	-
2	Panel Type 面板类型	IPS	-	-
3	Resolution 分辨率	720xRGBx720	Pixel	-
4	Display Mode 显示模式	Normally Black		-
5	Number of Colors 颜色数量	16.7M	-	-
6	Viewing Direction 使用视角	All	-	Note1
7	NTSC 色彩饱和度	68%	-	Typ.
8	Contrast Ratio 对比度	1000	-	
9	Luminance 亮度	-	cd/m2	Typ.
10	Module Size 模组尺寸	74.83(H)x78.98(V)x2.1(D)	mm	Note1
11	Panel Active Area 可视区域	71.93(H)x71.93(V)	mm	Note1
12	Pixel Pitch 像素尺寸	0.0999(H)x0.0999(V)	mm	-
13	Pixel Arrangement 像素排列	RGB-stripe		-
14	Weight 重量	TBD	g	-
15	Driver IC 驱动芯片	ST7703	-	-
16	Driver IC RAM Size 记忆体		bit	-
17	Light Source 背光源	10 White LEDs	-	-
18	Interface 接口方式	MIPI	-	-
19	Operating Temperature 工作温度	-20~+70	℃	-
20	Storage Temperature 存储温度	-30~+80	℃	-

Note 1: Please refer to the mechanical drawing；

注 1：请参照模组图；

2.Pin Assignments 接口定义

Pin No. Pin 序号	Symbol 符号	Function 功能描述
1	LEDK	Cathode of LED
2	LEDA	Anode of LED
3	GND	Ground
4	VCI 3.3	Analog Power Supply for LCM
5	VCI 3.3	Analog Power Supply for LCM
6	IOVCC	Logic Power Supply
7	RESET	LCM reset pin
8	TE	Tearing Effect pin.
9	GND	Ground
10	D3P	MIPI-DSI data Lane3 positive-end input
11	D3N	MIPI-DSI data Lane3 negtive-end input
12	GND	Ground
13	D2P	MIPI-DSI data Lane 2 positive-end input
14	D2N	MIPI-DSI data Lane 2 negtive-end input
15	GND	Ground
16	CLKP	MIPI-DSI clock Lane positive-end input
17	CLKN	MIPI-DSI clock Lane negtive-end input
18	GND	Ground
19	D1P	MIPI-DSI data Lane 1 positive-end input
20	D1N	MIPI-DSI data Lane1 negtive-end input
21	GND	Ground
22	D0P	MIPI-DSI data Lane 0 positive-end input/output pin
23	D0N	MIPI-DSI dataLane 0 negtive-end input/out putpin
24	GND	Ground
25	TP_VDD3.3V	Power Supply for Analog
26	TP_RESET	TP_The reset pin.
27	TP_INT	TP_The signal interrupts
28	TP_SDA	TP_Serial input/output data
29	TP_SCL	TP_Serial input CLK
30	TP_GND	TP_GND

3. Electrical Specification 电气特性

3.1Absolute Maximum Ratings 极限参数

Item	Symbol	Unit	Spec.		
				Typ.	Max.
Power Supply Voltage 1	IOVCC~VSSD	V	-0.3	-	+5.5
Power Supply Voltage 2	VCI ~ VSSA	V	-0.3	-	+6.6
Power Supply Voltage 3	VSP ~ VSSA	V	-0.3	-	+6.6
Power Supply Voltage 4	VSSA ~ VSN	V	-0.3	-	+6.6
Power Supply Voltage 5	VGH ~ VGL	V	-0.3	-	+35
Logic Input Voltage	V_{IN}	V	-0.3	-	IOVCC+0.3
Logic Output Voltage	Vo	V	-0.3	-	IOVCC+0.3
Differential Input Voltage	DSI_CP/DSI_CN DSI_D0P/DSI_D0P, DSI_D1P/DSI_D1N	V	-0.3	-	2.0

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3.2 Typical Operation Conditions 典型工作条件

Parameter	Symbol	Conditions	Spec.			Unit
			Min.	Typ.	Max.	
Power & Operating Voltages						
Logic Operating voltage	IOVCC	I/O supply voltage	1.65	1.8	2.0	V
Analog Operating voltage	VCI	Operation voltage	2.5	-	6.2	
Input / Output						
Logic High level input voltage	VIH	-	0.7IOVCC	-	IOVCC	V
Logic Low level input voltage	VIL	-	VSSD	-	0.3IOVCC	
Logic High level output voltage	VOH	IOH = -1.0mA	0.8IOVCC	-	IOVCC	
Logic Low level output voltage	VOL	IOL = +1.0mA	VSSD	-	0.2IOVCC	
Input leakage current	IIL	-	-1	-	1	μA
DC/DC Converter Operation						
VSP booster voltage	VSP	IVSP=1mA	4.5		6.2	V
VSN booster voltage	VSN	IVSN=-1mA	-6.2		-4.5	
VGH booster voltage	VGH	Ivgh=1mA	10		20	
VGL booster voltage	VGL	Ivgl=-1mA	-15		-7.5	
VGH and VGL difference	|VGH-VGL|	-	-	-	32	
Oscillator tolerance	OSC	25℃	-3		3	%
Source Driver						
Gamma reference voltage	VSPR	-	3.3		5.6	V
	VSNR	-	-5.6		-3.3	
Output voltage deviation	DVOS	VSSD+1.0 ~ VSPROUT-1.0	-	-	+/- 20	mV
		VSSD+0.1V ~ VSSD+1.0 VSPR-1.0 ~ VSPR-0.1V	-	-	+/- 50	mV
Output offset voltage	Voff	-		-	+/-50	mV
Standby Mode Current Consumption - 3 Power Mode						
Sleep In Mode	VSP	Ta=25℃ VSP=5.4V VSN=-5.4V IOVCC=1.8V	-	50	-	μA
	VSN		-	50	-	
	IOVCC		-	35	-	

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3.3 Backlight Circuit Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
LED Current	I_B		40	-	mA
LED Voltage	Vf		14-16		V
Power Consumption	P_{BL}	-		-	mW

3.4 LCD Current Consumption

Item	Symbol	Typ.	Max.	Unit
Full Mode	VCI	-	-	mA
Test Condition：VCI=2.8V，IOVCC=2.8V； Interface Driver type ： Row flip or column flip； TN Type=>All Black Pattern. IPS Type=>All White Pattern. Temperature：25℃.				
Sleep Mode	VCI	-	-	uA
Test Condition：VCI=2.8V，IOVCC=2.8V； DC/DC converter is enabled. Internal oscillator is started and panel scanning is started. Temperature：25℃.				

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4.Optical Specification 光学参数

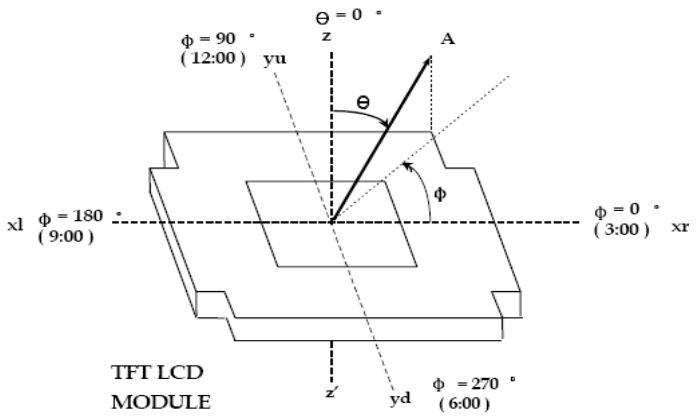
4.1 LCM Optical Characteristics 液晶模组光学特性

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle	Horizontal	Θ3	CR>10	80	85	-	°	Note 1
		Θ9		80	85	-	°	
	Vertical	Θ12		80	85	-	°	
		Θ6		80	85	-	°	
Contrast Ratio		CR	Θ= 0°	800	1000	-		Note 2
Luminance		cd/m2	Θ= 0°	-	-	-		
Uniformity		%	Θ= 0°	-	-	-		
Transmittance		T(%)	Θ= 0°	4.2	4.8	-		With APF
NSTC		%	Θ= 0°	63	68	-		
Reproduction Of color	Red	Rx	Θ= 0°	0.635	0.650	0.665		Note 3 *Color filter Glass(with out OC with C light)
		Ry		0.303	0.318	0.333		
	Green	Gx		0.248	0.263	0.278		
		Gy		0.550	0.565	0.580		
	Blue	Bx		0.125	0.140	0.155		
		By		0.071	0.086	0.101		
White		Wx	Θ= 0°	0.281	0.296	0.311		
		Wy		0.304	0.319	0.334		
Response Time		Tr+Tf	Θ= 0°	-	25	35	ms	Note 4

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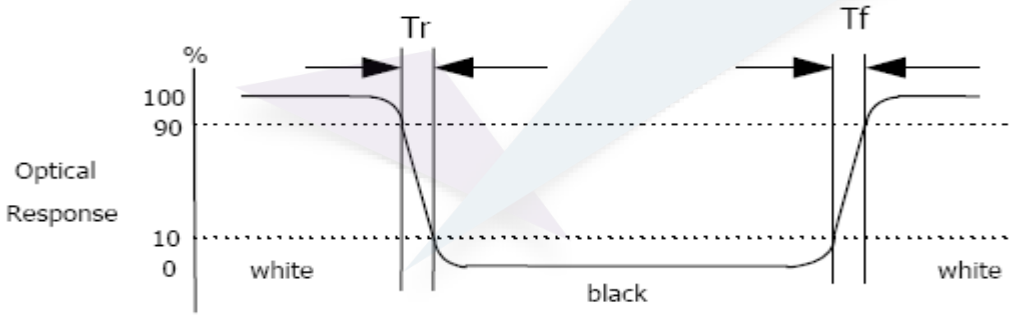
4.2 Measurement system 测量系统

4.2.1 LCM Viewing Angle



Viewing angle is the angle at which the contrast ratio is greater than 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface.

4.2.2 Response time



Response time is the time required for the display to transition from white to black (Rising time, Tr) and from black to white (Falling time, Tf) for additional information.

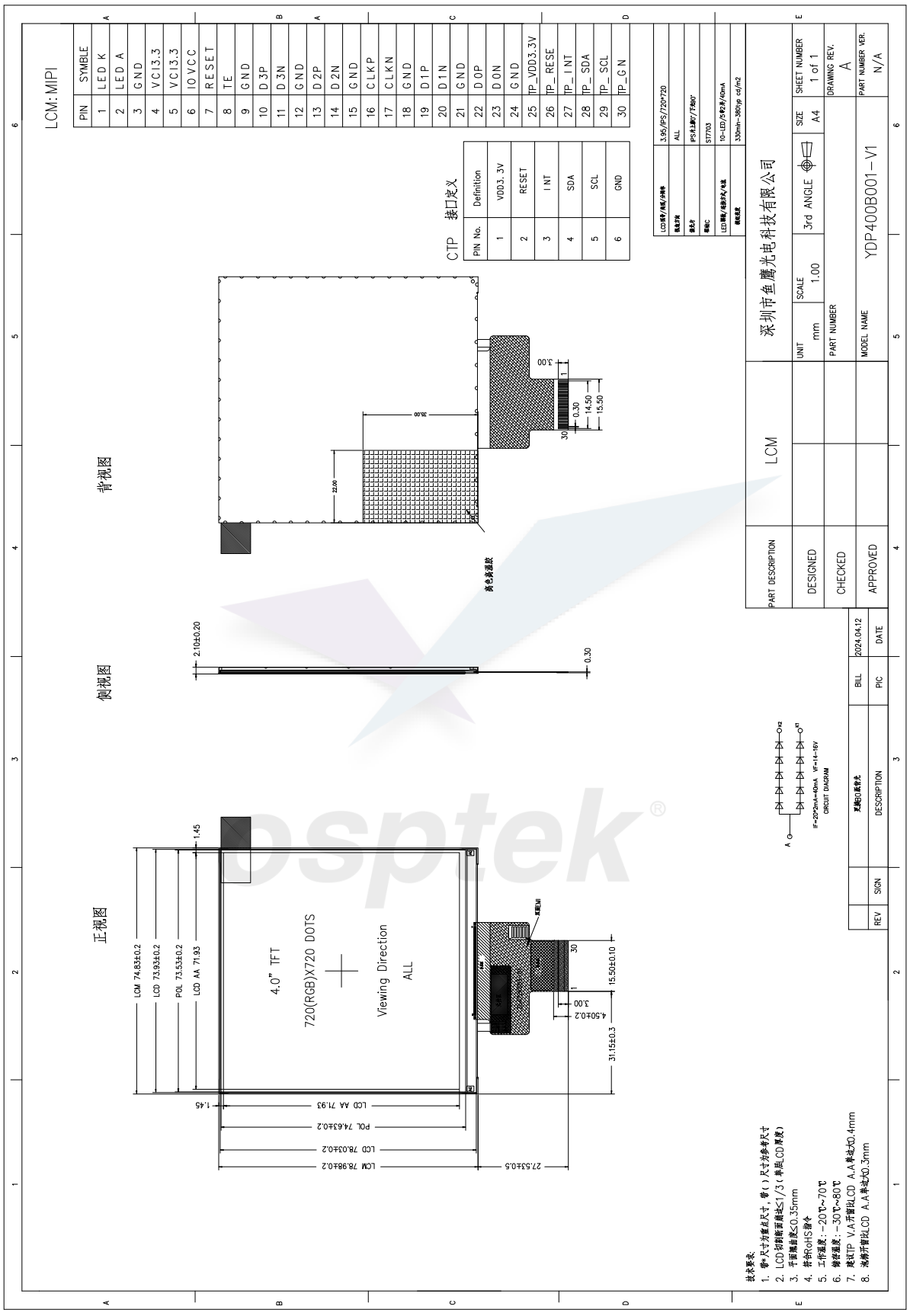
4.2.3 Contrast Ratio (CR)

Contrast Ratio (CR) is defined mathematically as:

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

Surface luminance is the center point across the LCD surface 500mm from the surface with all pixels displaying white.

5. Mechanical Drawing 模组图



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6. Reliability Test Items 可靠性测试项目

Test Item 测试项目	Test Condition 测试条件	Test result determinant gist 实验结果判定
High temperature storage 高温存储	85℃，24H；	Inspection after 2~4hours storage at room temperature, the sample shall be free from defects: 试验结束后,已测试的LCD样品必须在室内正常温湿度环境下放置2~4个小时以上才能进行功能和外观检查，样品不允许有以下缺陷： 1.Air bubble in the LCD; 模块中有气泡； 2.Non-display; 不显示； 3.Glass crack; 玻璃破碎； 4. The electrical characteristics requirements shall be satisfied. 需要满足模块电气性能。
Low temperature storage 低温存储	-30℃，24H；	
High temperature operation 高温运行测试	85℃，24H；	
Low temperature operation 低温运行测试	-30，24H；	
High temperature /humidity 高温高湿	60℃±3℃,90%±3%RH，24H；	
Thermal Shock 冷热冲击	-30℃/0.5h~+85℃/0.5h for a total 24 cycles；	
Vibration Test 振动测试	Frequency 10Hz~55Hz~10Hz Amplitude：1.5mm, X，Y，Z direction for total 1H；(Packing condition)	
ESD test 静电测试	±2KV, Human Body Mode, 150pF/330Ω； ±8KV, Air Mode, 150pF/330Ω；	

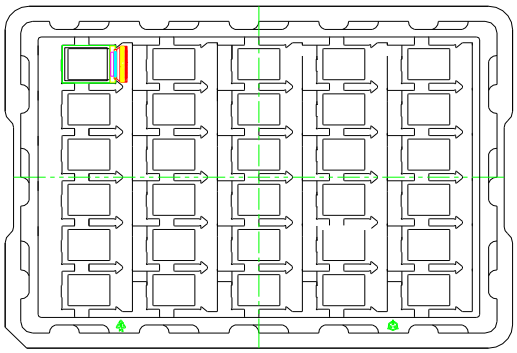
Remark: 注意：

1. The test samples should be applied to only one test item.
   每个被测试的模块只能用于其中的一个测试项目。
2. Sample size for each test item is 2pcs.
   每个测试项目的样品数量为2片。
3. Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.
   故障判断标准:基本规格,电气特性,机械特性,光电特性。

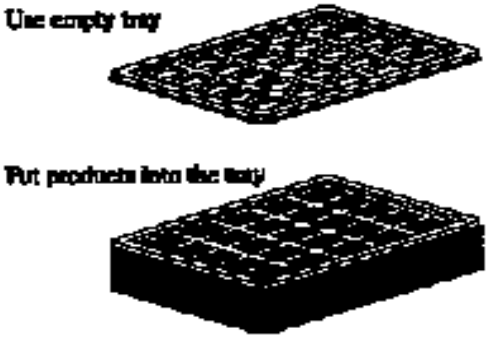
7. Packing and Storage Specification(Reference Only) 包装存储

7.1 Packing Method 包装方法

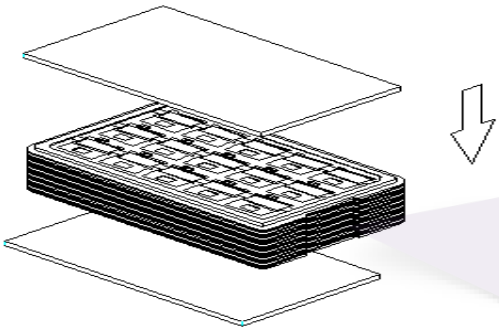
(1)



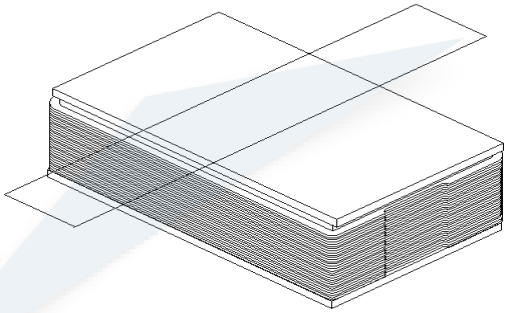
(2)



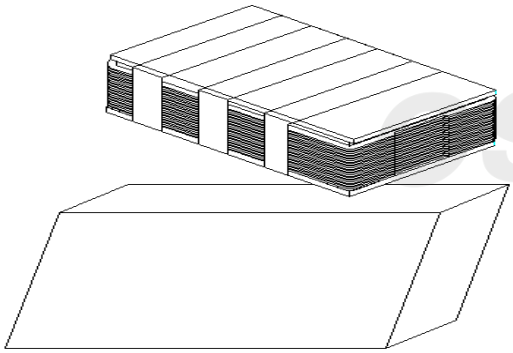
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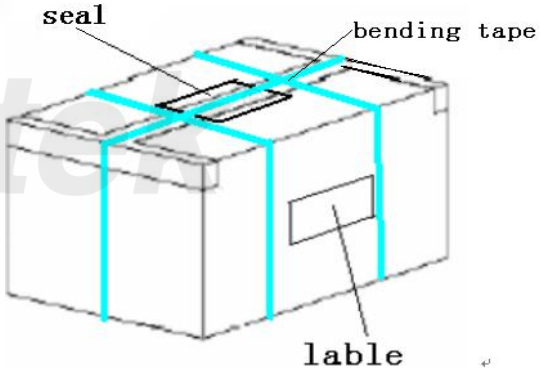
(4)



(5)



(6)



1. Put module into tray cavity. 把模块放进托盘.
2. Tray stacking. 托盘叠装.
3. Put 1 foam under the tray stack and 1 foam above. 在托盘上下放卡板.
4. Fix the cardboard to the tray stack with adhesive tape. 绑胶带.
5. Put the tray stack into carton. 把邦好的托盘放进纸箱.
6. Carton sealing with adhesive tape. 封纸箱.

7.2 Storage Method 存储方法

1. Store in an ambient temperature of 23°C±5°C, and in a relative humidity of 55%±15%. Don't exceed 12 months and expose to sunlight or fluorescent light.
   存储环境温度为 23±5°C，相对湿度为 55%±15%，存储不能超过 12 个月，不要长时间暴晒。
2. Store in a clean environment, free from dust, active gas, and solvent.
   存储在一个干净的环境，不受灰尘，活性气体和溶剂污染。
3. Store in antistatic container.
   存储在防静电环境。

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8. Announcements 注意事项

1. Do not attempt to disassemble or process the LCD module.
   请勿拆卸液晶显示模块。
2. Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached。
   不要在印制电路板上钻额外的孔，修改形状或更改印制线路板上元件的位置。
3. Except for soldering the interface, do not make any alterations or modifications with a soldering iron; Ensure welding temperature at 320 ° C to 350 ° C, the welding time control within the 10 s, welding note don't stay too long in the same place to avoid scald FPC.
   除焊接接口外，不要用烙铁做任何更改；焊接温度保证在 320°C-350°C，焊接时间控制在 10S 以内，焊接时注意不要在同一处停留时间太久以免烫伤 FPC。
4. Other matters in not clear before use, please contact our staff to guide.
   其他事项在不清楚使用之前，请联系我司人员指导进行。

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